The forward problem: hyperelastic membrane inflation

a constitutive ground truth for closed-surface tension (method M3)

Working notes (forward neo-Hookean membrane FE)

2026-06-20

Contents

1 Why a forward problem, and what it gives us

The tension-inference pipeline (companion notes, `tension_inference.tex`) solves the *inverse* membrane problem: given a deformed shape Γ and a pressure Δp, find the surface stress that balances them. On a closed surface this balance is statically determinate in principle but its discretisation carries a self-equilibrated null space, so the inverse solve is ill-posed and must be regularised. There is, consequently, no *exact* reference stress on a general closed shape against which to measure the inferred field — only analytic solutions for the sphere, spheroid and capsule.

The *forward* problem removes that gap. We posit a stress-free reference surface, a hyperelastic constitutive law and a pressure, and solve for the deformed configuration and its stress by energy minimisation. Because the elastic energy is convex for moderate strains, the forward solution is *unique*: the constitutive law and the reference together *select one member* of the self-equilibrated family that makes the inverse problem ambiguous. The forward stress is therefore a clean, null-mode-free ground truth — including the deviatoric and minor-principal components that the inverse solve recovers least well — and it is available on *arbitrary* shapes, not just the analytic ones. This is method M3 of the project matrix.

The key modelling choice (why the shape need not be prescribed exactly). We are interested in the *tension distribution*, not the precise deformed shape. On a pressurised membrane the tension is governed by shape and pressure (quasi-determinate), so it is insensitive to small shape changes. We exploit this: take the *target* shape as the (near) stress-free reference, choose a *stiff* material, and apply the pressure. The deformed shape then drifts from the target only by the elastic strain $\varepsilon \sim \sigma/(\text{membrane stiffness})$, which we make small ($\lesssim 1\%$) by choosing the shear modulus μ large. The recovered tension is then the equilibrium tension on (essentially) the target geometry. As μ grows the strain vanishes but the stress stays finite (it is fixed by equilibrium), so the limit is well behaved; μ is kept finite only so that the constitutive law genuinely picks the unique solution.

2 Kinematics of a surface membrane

Let the reference (stress-free) midsurface be $\Gamma_0 \subset \mathbb{R}^3$ with material coordinates $\boldsymbol{\xi} = (\xi^1, \xi^2)$ and embedding $\boldsymbol{X}(\boldsymbol{\xi})$. Deformation maps it to Γ with $\boldsymbol{x}(\boldsymbol{\xi})$. The reference and deformed covariant tangent bases are

$$\boldsymbol{A}_\alpha = \frac{\partial \boldsymbol{X}}{\partial \xi^\alpha}, \qquad \boldsymbol{a}_\alpha = \frac{\partial \boldsymbol{x}}{\partial \xi^\alpha}, \qquad \alpha \in \{1, 2\}. \tag{1}$$

The *surface deformation gradient* is the $(3{\times}2)$ two-point tensor

$$\boldsymbol{F} = \boldsymbol{a}_\alpha \otimes \boldsymbol{A}^\alpha, \qquad \boldsymbol{A}^\alpha \cdot \boldsymbol{A}_\beta = \delta^\alpha_\beta, \tag{2}$$

mapping reference tangent vectors to deformed ones. The $(2{\times}2$, in-plane) right Cauchy–Green tensor and its invariants are

$$\boldsymbol{C} = \boldsymbol{F}^\top \boldsymbol{F}, \qquad C_{\alpha\beta} = \boldsymbol{a}_\alpha \cdot \boldsymbol{a}_\beta, \qquad I = \mathrm{tr}_A\, \boldsymbol{C} = A^{\alpha\beta} C_{\alpha\beta}, \qquad J = \sqrt{\det\bigl(A^{\alpha\gamma} C_{\gamma\beta}\bigr)}, \tag{3}$$

where $A^{\alpha\beta}$ is the inverse reference metric. I is the trace of the in-plane stretch ($I = \lambda_1^2 + \lambda_2^2$ in principal stretches λ_i) and $J = \lambda_1 \lambda_2$ is the local *areal* stretch (deformed area / reference area). For a flat, isotropically-meshed reference one may take $A_{\alpha\beta} = \delta_{\alpha\beta}$ per element (Section 5).

3 Constitutive law: 2D membrane neo-Hookean

We use the same membrane neo-Hookean energy as the cMSM forward simulations (Marín-Llauradó et al., 2023), a strain energy *per unit reference area*

$$\psi(I, J) = \mu\,(I - 2) + \lambda\,(J - 1)^2, \tag{4}$$

with shear modulus μ and an areal-penalty modulus λ (their choice $\lambda = 2\mu$; both have units of surface tension, N/m). The $\mu(I-2)$ term resists shear/stretch; the $\lambda(J-1)^2$ term resists area change and supplies a stable equilibrium under pressure. The second Piola–Kirchhoff surface stress follows from $\boldsymbol{S} = 2\,\partial\psi/\partial\boldsymbol{C}$:

$$\boldsymbol{S} = 2\frac{\partial \psi}{\partial I}\,\boldsymbol{A}^{-1} + 2\frac{\partial \psi}{\partial J}\,\frac{\partial J}{\partial \boldsymbol{C}} = 2\mu\,\boldsymbol{A}^{-1} + \lambda\,(J-1)\,J\,\boldsymbol{C}^{-1}, \tag{5}$$

using $\partial I/\partial \boldsymbol{C} = \boldsymbol{A}^{-1}$ (reference inverse metric) and $\partial J/\partial \boldsymbol{C} = \frac{1}{2} J\, \boldsymbol{C}^{-1}$. The physical *Cauchy membrane stress* (force per unit deformed length, i.e. the surface tension tensor compared against the inference) is the push-forward

$$\boldsymbol{\sigma} = \frac{1}{J}\, \boldsymbol{F}\, \boldsymbol{S}\, \boldsymbol{F}^{\top}, \tag{6}$$

a symmetric in-plane (3×3, rank-2) tensor with $\boldsymbol{\sigma}\, \boldsymbol{n} = \boldsymbol{0}$. Its non-zero eigenvalues are the principal tensions σ_1, σ_2 that we compare to the inferred field. (Note σ here is a force-per-length *resultant*, matching our N; divide by thickness t if a true stress is wanted.)

4 Equilibrium: energy functional and weak form

4.1 Total potential energy

The deformed configuration minimises the total potential energy

$$\Pi[\boldsymbol{x}] = \underbrace{\int_{\Gamma_0} \psi(I, J)\, \mathrm{d}A_0}_{\text{elastic energy}} \; - \; \underbrace{\Delta p\, V[\boldsymbol{x}]}_{\text{pressure work}}, \qquad V[\boldsymbol{x}] = \frac{1}{3} \oint_{\Gamma} \boldsymbol{x} \cdot \boldsymbol{n}\, \mathrm{d}a, \tag{7}$$

where V is the volume enclosed by the closed deformed surface (divergence theorem) and Δp is the internal–external pressure difference. Stationarity $\delta\Pi = 0$ for all admissible variations $\delta\boldsymbol{x}$ gives the weak form

$$\underbrace{\int_{\Gamma_0} \boldsymbol{S} : \delta\boldsymbol{E}\, \mathrm{d}A_0}_{\text{internal virtual work}} \; = \; \underbrace{\Delta p \oint_{\Gamma} (\boldsymbol{n} \cdot \delta\boldsymbol{x})\, \mathrm{d}a}_{\text{external (pressure) virtual work}}, \qquad \delta\boldsymbol{E} = \tfrac{1}{2}\big(\boldsymbol{F}^{\top} \delta\boldsymbol{F} + \delta\boldsymbol{F}^{\top} \boldsymbol{F}\big), \tag{8}$$

$\boldsymbol{E} = \frac{1}{2}(\boldsymbol{C} - \boldsymbol{A})$ being the Green–Lagrange strain. The right-hand side is a *follower* load: the normal $\boldsymbol{n}$ and area element $\mathrm{d}a$ are evaluated on the *deformed* surface, so the pressure term depends on the unknown $\boldsymbol{x}$. Equivalently, the pressure work is simply $\Delta p\, \delta V = \Delta p \oint \boldsymbol{n} \cdot \delta\boldsymbol{x}\, \mathrm{d}a$, which is where the follower form comes from.

4.2 Pressure control vs. volume control

Two ways to drive the inflation:

- **Pressure-controlled:** Δp prescribed, V free. Simple, but a closed membrane can be *unstable* once past the inflation limit point (the balloon instability: $\mathrm{d}(\Delta p)/\mathrm{d}V < 0$), where Newton diverges.
- **Volume-controlled:** prescribe the enclosed volume $V = V^{\star}$ via a Lagrange multiplier; the multiplier *is* the pressure Δp (a readout). The augmented stationarity is $\int \boldsymbol{S} : \delta\boldsymbol{E}\, \mathrm{d}A_0 - \Delta p\, \delta V = 0$, $V[\boldsymbol{x}] = V^{\star}$. This traverses the limit point stably and is exactly the device the cMSM vertex model used to read off lumen pressure.

For a stiff membrane at modest Δp we are far below the limit point and pressure control is adequate; volume control is the robust fallback.

5 Finite-element discretisation: P1 membrane triangles

The reference surface is a triangulation; the unknowns are the deformed nodal positions $\boldsymbol{x}_i \in \mathbb{R}^3$ (3 DOF/node). We use the **constant-strain triangle (CST) membrane element**: piecewise-linear (P1) geometry, so each triangle has a *constant* deformation gradient and hence constant stress. This is the standard membrane element and the counterpart of the P1 elements in the inverse solver (`getShapeFunctions.m`, `'Triangle'` in the cMSM code).

Element deformation gradient. On a triangle with reference vertices $\boldsymbol{X}_0, \boldsymbol{X}_1, \boldsymbol{X}_2$ and deformed vertices $\boldsymbol{x}_0, \boldsymbol{x}_1, \boldsymbol{x}_2$, form the edge matrices

$$\boldsymbol{D}_X = [\, \boldsymbol{X}_1 - \boldsymbol{X}_0 \;\; \boldsymbol{X}_2 - \boldsymbol{X}_0 \,] \in \mathbb{R}^{3\times 2}, \qquad \boldsymbol{D}_x = [\, \boldsymbol{x}_1 - \boldsymbol{x}_0 \;\; \boldsymbol{x}_2 - \boldsymbol{x}_0 \,] \in \mathbb{R}^{3\times 2}, \tag{9}$$

and a 2D reference frame (orthonormal tangent basis $\boldsymbol{t}_1, \boldsymbol{t}_2$ of the reference triangle) so that $\hat{\boldsymbol{D}}_X = [\boldsymbol{t}_1\, \boldsymbol{t}_2]^\top \boldsymbol{D}_X \in \mathbb{R}^{2\times 2}$ is invertible. Then the surface deformation gradient and Cauchy–Green follow as

$$\boldsymbol{F} = \boldsymbol{D}_x\, \hat{\boldsymbol{D}}_X^{-1}\, [\boldsymbol{t}_1\, \boldsymbol{t}_2]^\top, \qquad \boldsymbol{C} = \boldsymbol{F}^\top \boldsymbol{F}, \qquad A_0 = \tfrac{1}{2}\big|\boldsymbol{D}_X(:,1) \times \boldsymbol{D}_X(:,2)\big|, \tag{10}$$

with A_0 the reference triangle area. $I, J, \boldsymbol{S}, \boldsymbol{\sigma}$ are then constant on the element via (5)–(6).

Element internal force and tangent. The element energy is $\Psi_e = A_0\, \psi(I, J)$. Its gradient and Hessian with respect to the 9 element DOFs $\boldsymbol{u}_e = (\boldsymbol{x}_0, \boldsymbol{x}_1, \boldsymbol{x}_2)$ give the internal force and material+geometric tangent

$$\boldsymbol{f}_e^{\text{int}} = \frac{\partial \Psi_e}{\partial \boldsymbol{u}_e}, \qquad \boldsymbol{K}_e^{\text{mat+geo}} = \frac{\partial^2 \Psi_e}{\partial \boldsymbol{u}_e\, \partial \boldsymbol{u}_e}, \tag{11}$$

assembled into the global residual and stiffness. These derivatives are available in closed form for (4) but are most safely obtained by *automatic differentiation* of $\Psi_e(\boldsymbol{u}_e)$ (Section 9).

Pressure (follower) load. The discrete enclosed volume and its gradient (the pressure force) are

$$V = \frac{1}{6} \sum_{\text{tri}} \boldsymbol{x}_0 \cdot (\boldsymbol{x}_1 \times \boldsymbol{x}_2), \qquad \boldsymbol{f}^{\text{ext}} = \Delta p\, \frac{\partial V}{\partial \boldsymbol{x}}, \tag{12}$$

which distributes the pressure as nodal forces along deformed normals. Its derivative $\partial^2 V / \partial \boldsymbol{x}\, \partial \boldsymbol{x}$ is the *load (pressure) stiffness*; it is non-symmetric, so the global tangent $\boldsymbol{K} = \boldsymbol{K}^{\text{mat+geo}} - \Delta p\, \partial^2 V / \partial \boldsymbol{x}^2$ is non-symmetric and must be solved with a general (not Cholesky) sparse solver.

6 Boundary conditions: a free-floating pressurised body

The target case is a *closed, free-floating* surface (embryo) under internal pressure — *no* physical boundary, unlike the pinned open domes of cMSM. This is admissible because a closed surface under uniform pressure is *self-equilibrated*: the net pressure force $\oint \Delta p\, \boldsymbol{n}\, \mathrm{d}a$ and torque vanish. The only obstruction is the 6-dimensional **rigid-body null space** (3 translations + 3 rotations) of the stiffness matrix. Two standard cures:

- **Minimal "3–2–1" restraint:** fix one node in all 3 directions, a second in 2, a third in 1 (6 constraints total, nodes non-collinear). Because the body is in self-equilibrium the reactions are essentially zero, so this introduces *no spurious stress* — it only removes the rigid modes.
- **Rigid-mode projection:** keep all DOFs free and project the 6 rigid modes out of the residual and increment (or use a minimum-norm `lsqr` solve). Equivalent stress, no anchor nodes.

The pressure load is consistent (orthogonal to the rigid modes), so a solution exists. Volume control (Section 4) adds one Lagrange-multiplier row/column to the tangent and replaces Δp by an unknown; the rigid-mode treatment is unchanged.

7 Solution algorithm

1. **Mesh & reference.** Build the reference surface (`vedo.IcoSphere`, the capsule builder, or an embryo mesh); store reference frames, edge matrices $\hat{\boldsymbol{D}}_X$ and areas A_0 once.
2. **Load stepping.** Ramp Δp (or $V^\star$) in n_{step} increments from 0 to the target. For a stiff membrane the response is nearly linear, so $n_{\text{step}} = 1$–5 suffices; use more only for soft materials or near the limit point.
3. **Newton iteration** per step: assemble residual $\boldsymbol{r} = \boldsymbol{f}^{\text{int}} - \Delta p\, \partial V/\partial \boldsymbol{x}$ and tangent $\boldsymbol{K}$; apply the 3–2–1 constraints (or projection); solve $\boldsymbol{K}\,\delta\boldsymbol{u} = -\boldsymbol{r}$ with a sparse direct solver (`scipy.sparse.linalg.spsolve`) or GMRES for large/non-symmetric systems; update $\boldsymbol{x} \leftarrow \boldsymbol{x} + \delta\boldsymbol{u}$; iterate to $\|\boldsymbol{r}\| < \text{tol}$ (typically 3–6 iterations).
4. **Stress recovery.** From the converged $\boldsymbol{F}$ per element compute $\boldsymbol{\sigma}$ (6) and its eigenvalues σ_1, σ_2; average to nodes if a nodal field is wanted. Export to VTK alongside the inferred field.
5. **Diagnostics.** Track the shape drift $\max \|\boldsymbol{x} - \boldsymbol{X}\|/R$ (should be $\lesssim 1\%$ for the stiff-reference strategy), the enclosed volume, and the smallest tangent eigenvalue (limit-point proximity).

8 Validation plan

Following the project rule (sphere first):

- **Sphere.** A neo-Hookean spherical membrane inflates self-similarly with a closed-form pressure–stretch relation $\Delta p(\lambda)$; check the solver reproduces it and that $\sigma_1 = \sigma_2 = \Delta p\, R/2$ (resultant). Verify the balloon limit point is *not* crossed at the working Δp.
- **Prolate / oblate spheroid, capsule.** Take the analytic shape as the stress-free reference, inflate with stiff μ, confirm the shape drift is small and the recovered σ_1, σ_2 match the axisymmetric closed forms (Section 1); these become the M3 ground truth.
- **Cross-check against inference.** Feed the (slightly) deformed shape to the inverse FEM and compare — the forward field is the unique, null-mode-free target the inference should recover.

9 Software / packages

The forward problem is genuinely nonlinear (hyperelastic energy + follower load + Newton), unlike the linear inverse solve. Three viable stacks, in order of fit to this project:

(A) Hand-rolled `scipy.sparse` + automatic differentiation (recommended). Consistent with the project's no-framework convention and reuses the existing `vedo` mesh/IO and VTK export. Needed packages:

- `numpy`, `scipy` — array maths; `scipy.sparse` + `scipy.sparse.linalg` (`spsolve`/`gmres`) for the Newton linear solves; `scipy.optimize` optionally for an energy-minimisation formulation (trust-region / L-BFGS as a robust fallback to Newton).
- **Automatic differentiation** of the element energy $\Psi_e(\boldsymbol{u}_e)$ to get exact internal forces and tangents without hand-deriving (5) and the geometric/pressure stiffness. Either `jax` (`jax.grad`, `jax.hessian`, `vmap` over elements — fastest, vectorised) or `autograd`. This is the single most labour-saving dependency; the CST energy is small (9 DOF/element) so per-element AD is cheap.

- `vedo` (already installed) — reference-mesh construction and VTK output; `numpy`-side assembly into global sparse matrices.

Estimated ∼300–500 lines, fully transparent, no Christoffel/shell machinery.

(B) `scikit-fem` (installed, 12.0.1). Can assemble the forms and apply constraints, and supports custom nonlinear (bi)linear forms with a Newton loop. However, membrane elements *embedded in* $\mathbb{R}^3$ (a 2-manifold element carrying 3 nodal DOFs, with the deformed metric as unknown) are not its native setting; the follower-pressure load and the deformed-configuration geometry would still be hand-coded. Usable for the assembly bookkeeping, but it does not remove the hyperelastic-membrane-specific work.

(C) `FEniCS/FEniCSx` or `Firedrake`. The most capable for nonlinear hyperelastic shells/membranes: UFL expresses ψ, $\boldsymbol{F}$, the follower load and the Lagrange-multiplier volume constraint symbolically, with automatic tangents and robust nonlinear solvers (PETSc SNES). The cost is a heavy dependency that is awkward to install on Windows (typically via `conda`/WSL/Docker) and a departure from the project's lightweight stack. Best reserved for if the hand-rolled solver proves limiting (e.g. large embryo meshes needing PETSc scalability).

Recommendation. Start with (A): `numpy` + `scipy.sparse` for assembly/solves, `jax` for element-level autodiff of (4), and `vedo` for meshes and VTK. It matches the existing pipeline, keeps the whole forward solver inspectable, and is more than sufficient for sphere/spheroid/capsule and embryo-scale meshes. Escalate to (C) only if PETSc-level performance becomes necessary.

Summary

The forward problem inflates a stress-free reference surface under (follower) internal pressure using a P1 CST membrane neo-Hookean model (4), minimising the total potential energy (7) by Newton iteration with a 3–2–1 (or projection) treatment of the rigid-body modes of a free-floating closed body. Using the target shape as a *stiff* reference keeps the deformed shape within $\sim 1\%$ of the target, so the recovered σ_1, σ_2 are the equilibrium tensions on (essentially) the desired geometry — a unique, null-mode-free ground truth (M3) for arbitrary closed surfaces, including embryos with no analytic solution. The recommended implementation is hand-rolled `numpy/scipy.sparse` with `jax` autodiff for the element tangents and `vedo` for mesh I/O.